Final Project Report:

Forecasting Pollution Using Time Series Analysis

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Abstract

This project explores the use of time series modeling to analyze and forecast the PM2.5 pollution levels when the air quality in Beijing is at its worst. The data is collected hourly, but to make it more manageable and representative, I eliminate the size to three key time points at 0:00, 8:00, and 16:00 per day. This project uses the Box-Jenkins approach and the unit roots tests. In the Box-Jenkins approach, I built an ARIMA model after log-transformation. In the unit roots tests, I applied Dickey-Fuller (DF), Augmented Dickey-Fuller (ADF), and Phillips-Perron (PP) tests, all of which confirmed stationarity.

1. Introduction

Air pollution is a major environmental issue affecting human health. I remember that when I was in elementary school, it was the first time I knew that there was a thing called PM2.5, which was the main source causing the severe haze at that time. Also, it was the first time that I knew that the world was covered by haze but not the fog. From then on, the government of China implemented many policies to try to ease this situation and create good air quality. However, this cannot be solved rapidly, just like what London faced back in 1952. Therefore, we all care a lot about PM2.5 and we can see that the news every day will report the concentration of PM2.5 for each city, and it suggests those who are in severely polluted cities to wear a mask. The situation of bad air quality happened for a long time, and I cannot even remember what it looks like by having clouds in the day and stars in the night. Due to the poor air quality, I had rhinitis and conjunctivitis due to allergies. The air quality becomes better when the pandemic comes. Everything has two sides, and one of the good sides of the pandemic is that it makes the air quality much better.

As a result, I want to analyze and see what happens to the PM2.5 pollution levels when the air quality in Beijing is at its worst, and I want to see whether it becomes better or worse around the time period of 2010-2014. Past studies on the datasets have applied a wide range of time series and machine learning methods. The ARIMA and SARIMA models have been used frequently. This dataset explored deep learning models such as Long Short-Term Memory (LSTM) networks for improved forecasting accuracy. The data is collected hourly, but to make it more manageable and representative, I eliminate the size to three key time points at 0:00, 8:00, and 16:00 per day. The methods I use for this project include log transformation, ARIMA, and SARIMA modeling, estimation on the parameters, forecasting, and unit roots test which I applied Dickey-Fuller (DF), Augmented Dickey-Fuller (ADF), and Phillips-Perron (PP) tests.

After analysis, we can see that the log-transformed data is stationary, which has been confirmed by the unit roots test. In addition, the ARIMA (3,0,2) model provided reasonably accurate short-term forecasts of PM2.5 pollution levels. Also, the forecast exhibits a gradual upward trend during forecast periods, which may reflect persistent accumulation under certain weather conditions.

2. Data

The dataset used in this project comes from Kaggle and is titled "Air Pollution Forecasting". It reports hourly weather and air pollution data collected at the U.S. Embassy in Beijing over a five-year period and has been uploaded by Rupak Roy. The dataset is widely used for time series modeling and machine learning, particularly for air quality forecasting using both univariate and multivariate approaches.

This dataset covers the time period from 2010 to 2014, with each row representing an hourly observation. To make the analysis more manageable and to better reflect daily pollution patterns, I selected three key time points per day: 00:00, 08:00, and 16:00.

The dataset contains the following variables:

- pollution: PM2.5 concentration level, the target variable.
- dew: Dew point temperature (°C).
- temp: Temperature (°C).
- press: Atmospheric pressure (hPa).
- wnd_dir: Wind direction (character type).
- wnd_spd: Wind speed (m/s).
- snow: Cumulative snow hours.
- rain: Cumulative rain hours.
- pollution_log: Log-transformed version of pollution for stationarity testing.

The total number of observations after filtering is 5475, and the data is numerical except for wind direction and date, which are in character format.

The reason I select this data is that I want to use methods in time series to see what happened during the time that China was experiencing a severe air pollution. By analyzing pollution trends and weather patterns from this dataset, I hope I can uncover methods to improve the forecasting accuracy and address on the awareness of protecting the environment.

3. Methodology

This project adopts multiple approaches to time series modeling like the Box-Jenkins approach (SARIMA (p, d, q) × (P, D, Q)), Unit Root tests, and ARMAX Model. Both ARIMA and ARMAX models were used for short-term forecasting.

3.1 Box-Jenkins Approach

The primary modeling framework used is the ARIMA model and SARIMA, denoted as SARIMA (p, d, q) × (P, D, Q). The steps followed in this project are:

- Step 1: Plot of the time series data.
- Step 2: Log transformation of the time series data.
- Step 3: Identification of the difference and the first implementation of the unit root test.
- Step 4: Estimation of parameters.
- Step 5: Diagnostics.
- Step 6: Model Selection.

3.2 Unit Root Tests

To determine whether the time series is stationary, three types of unit root tests were applied:

- Dickey-Fuller (DF) Test: A basic test checking for non-stationarity without lags.
- Augmented Dickey-Fuller (ADF) Test: Extends DF by including lagged differences to account for autocorrelation.
- Phillips-Perron (PP) Test: Adjusts for serial correlation and heteroskedasticity in the errors.

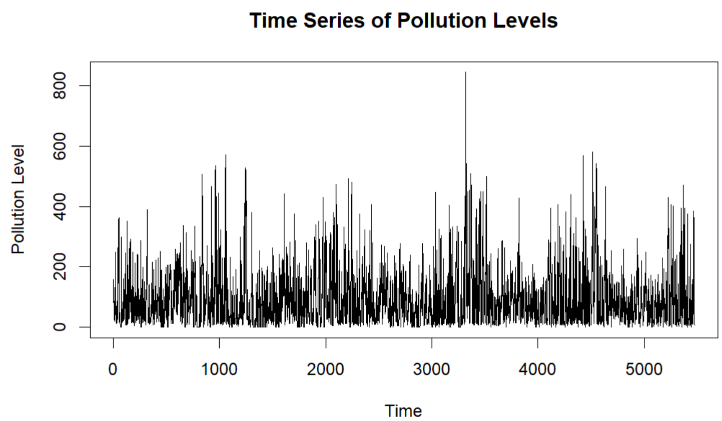
3.3 ARMAX Model

To analyze the exogenous meteorological variables: temperature, dew point, and pressure, this project uses the ARMAX model by first fitting the model and then do the lag order selection. Additionally, the diagnostic tests were performed to validate the model.

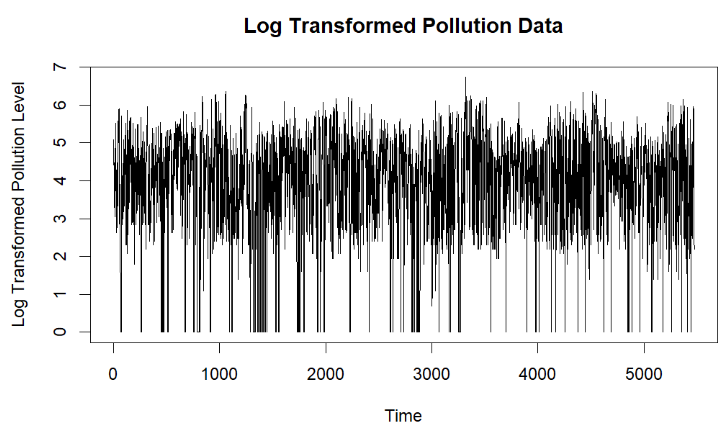
4. Results

4.1 Results from Box-Jenkins approach (SARIMA (p, d, q) × (P, D, Q))

From *graph 1*, we can see the plot of the time series of PM2.5 pollution levels from the Beijing dataset, sampled three times per day (00:00, 08:00, and 16:00) across five years. The data shows highly volatile fluctuations, and some extreme peaks suggest potential irregularities and non-stationary with visible trends and variance instability.



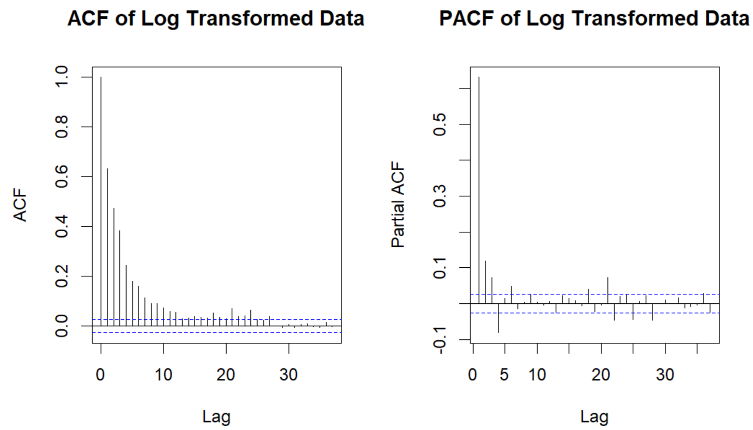
Graph 1 Plot of time series of PM2.5 pollution levels



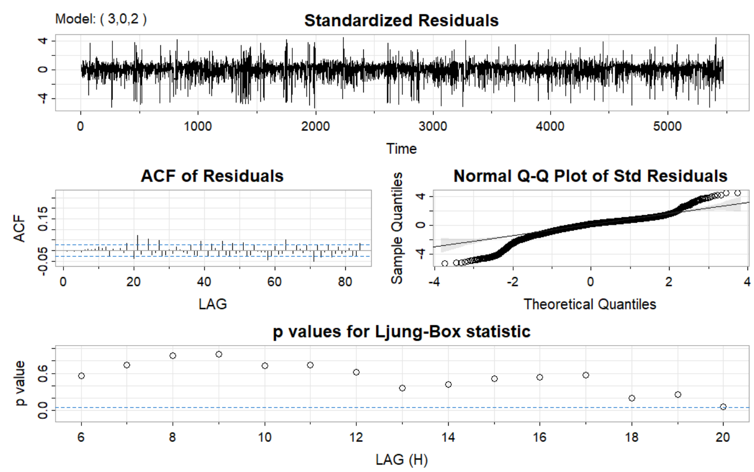
Graph 2 Plot of Log-transformed Pollution Data.

Therefore, I applied log transformation to stabilize the variance and make the data more suitable for ARIMA modeling. *Graph 2* appears more stable in terms of variance, though stationarity still needs to be tested.

Consequently, I generate the ACF and PACF graph to analyze if it is stationary. From *graph 3*, we can see that the ACF is decreasing gradually and the PACF sharply drops to around the interval of confidence after the first few lags. Even though the graph indicates that some of the lags are irregular to be stationary, the ADF test results support the stationarity of the log-transformed data, as the p-value is 0.01, which is less than 0.05 and confirms that no further differencing is needed.



Graph 3 Plot of ACF/PACF of Log Transformed Data

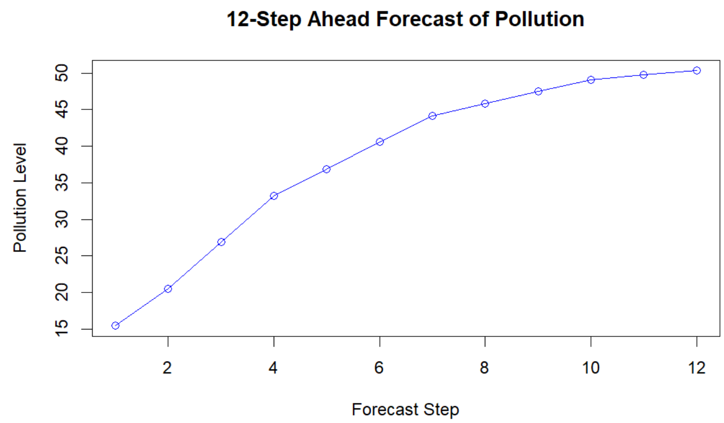


Graph 4 Diagnostic Plots of SARIMA (3,0,2) Model

By confirming its stationarity, I generate a fitted SARIMA (3,0,2) model (*graph 4*). The standardized residuals show no apparent patterns, and most values lie within a range typically associated with ±3 standard deviations, suggesting no extreme outliers or structural inadequacy. The plot of the estimated ACF of the residuals shows nearly no apparent departure from the model assumption, except a few lags. The normal Q-Q plot of the residuals show that the assumption of normality is reasonable, although there are some outliers. From the p values for Ljung-Box statistics, it shows that the Q-statistics are not statistically significant.

The last step of the Box-Jenkins approach is to select the best model. I use the code of "auto.arima()" to find the best model with the lowest Bayesian information criterion (BIC), which is the model I applied previously: SARIMA (3, 0, 2) model or ARIMA (3, 0, 2) model, which is nonseasonal.

After analyzing, I generated the 12-step ahead forecast of PM2.5 pollution levels using the SARIMA (3,0,2) model (*graph 5*). The forecast shows a smooth and steady upward trend, suggesting that the pollution level is likely to gradually increase over the next 12 steps. This rise might indicate that in the next few days, there will be a short-term accumulation of pollutants, which can tell the public pay attention to the potential bad air quality in the future. However, if we see the pollution levels at these times, they are all less than fifty, which is not high and shows that the air quality is not that bad.



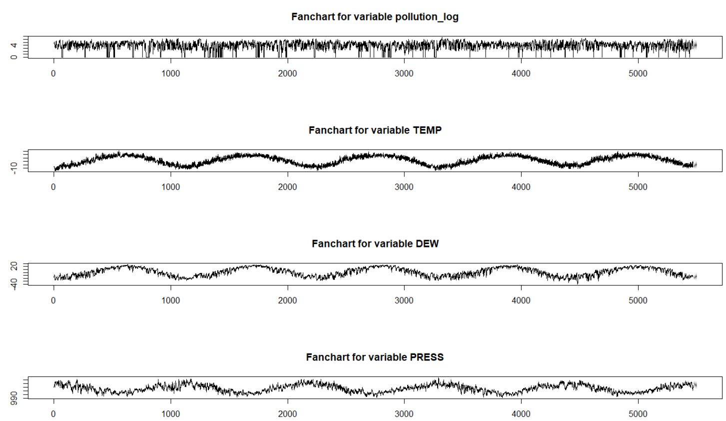
Graph 5 Plot of 12-Step Ahead Forecast of Pollution

4.2 Unit Root Tests

To confirm that the log-transformed PM2.5 pollution series is stationary, I applied three standard unit root tests: the Dickey-Fuller (DF) test, the Augmented Dickey-Fuller (ADF) test, and the Phillips-Perron (PP) test. All three tests support the stationary of the series. The DF test produced a test statistic of -35.097 with a p-value less than 0.01, strongly rejecting the null hypothesis of a unit root. The ADF test, with lag order 17, gave test statistics of -14.066 and a p-value of 0.01, confirming stationarity. The PP test returned a Z(alpha) statistic of -2257.2 and a p-value of 0.01, further rejecting the unit root null. For detailed coding and results, please refer to the appendix.

4.3 ARMAX Model

In order to further enhance forecasting performance and account for the influence of meteorological variables on PM2.5 levels, this project applied an ARMAX model using temperature, dew point, and pressure as exogenous variables. Although the selection criteria (AIC, HQ, SC, and FPE) all suggested a lag order of 10, I used a simplified ARMAX model with lag = 2 to make the value more manageable and clearer to analyze (a full model with lag = 10 was also estimated and gave similar conclusions). The results for the pollution equation indicate that both lag-1 and lag-2 pollution values were statistically significant predictors since the p-value for the estimators are all less than 0.01, confirming strong autocorrelation in pollution levels. Moreover, all three other variables (temperature, dew point, and air pressure) all result in significant association with the pollution level with p less than 0.05. The multiple R-squared value for the pollution equation was 0.4416, indicating that the model explains approximately 44% of the variance in log-transformed pollution levels. This indicates that the model explained some of the variance but did not fit very well. But estimation results for the other variables similarly showed statistically significant relationships with past values of pollution and other exogenous variables, with adjusted R-squared values ranging from about 91% to 93%, reflecting a good fit in the multivariate system. For the detailed results and coding, please refer to the appendix.



Graph 6 Plot of Fancharts for Variables in the Model

To visualize the model fit, I plotted fancharts for all variables in the ARMAX model (*graph 5*). The forecast of the log transformation of pollution remains relatively stable with slight fluctuations and narrow confidence intervals, suggesting the model captures the short-term variation effectively. Both variables of temperature (TEMP) and dew point (DEW) show clear cyclical trends, likely reflecting daily or seasonal temperature and dew point patterns, and they all visualized stationary. The fitted lines follow the actual patterns closely, indicating the model's accuracy in capturing these meteorological factors. Pressure (PRESS) also displays a gentle cyclical structure. The fitted curve tracks the general trend well, although some sharp deviations exist. Besides, the graph also shows that the pressure is negatively correlated with temperature and dew point.

Conclusion and Future Study

In this project, the time series modeling approach was applied to analyze the PM2.5 pollution level in Beijing. After pre-processing and log-transforming the raw data, I confirmed the smoothness of the data through ADF and PP unit root tests. A SARIMA (3,0,2) model was then fitted, which provided reasonable short-term predictions and showed a gradual increase in pollution levels.

In addition to the SARIMA model, I applied the ARMAX model to consider meteorological variables such as temperature, dew point, and barometric pressure. The results show that these variables are significant predictors of pollution levels, especially dew point and lagged pollution values, which are prominent in the model.

An important finding of this study is that both past pollution levels and current weather conditions play a key role in predicting PM2.5 concentrations. This also reflects the reality of environmental changes characterized by the fact that pollution tends to build up gradually under specific meteorological conditions.

Going forward, subsequent studies could further explore the influence of seasonal factors, integrate more external variables such as traffic flow, and compare traditional statistical models with machine learning methods (e.g., LSTM) to improve prediction accuracy. This study demonstrates the potential of combining statistical modeling with domain knowledge to contribute to a deeper understanding and more effective management of urban air pollution problems.

References

Kawahara, T., & Fujikoshi, Y. (2016). Introduction to Time Series Modeling with Applications in R. Springer. https://doi.org/10.1007/978-3-319-52452-8

Roy, R. (n.d.). LSTM datasets (multivariate & univariate) [Data set]. Kaggle. Retrieved March 21, 2025, from https://www.kaggle.com/datasets/rupakroy/lstm-datasets-multivariate-univariate/data

Appendix

See the following pages for the complete R Markdown output and related results.

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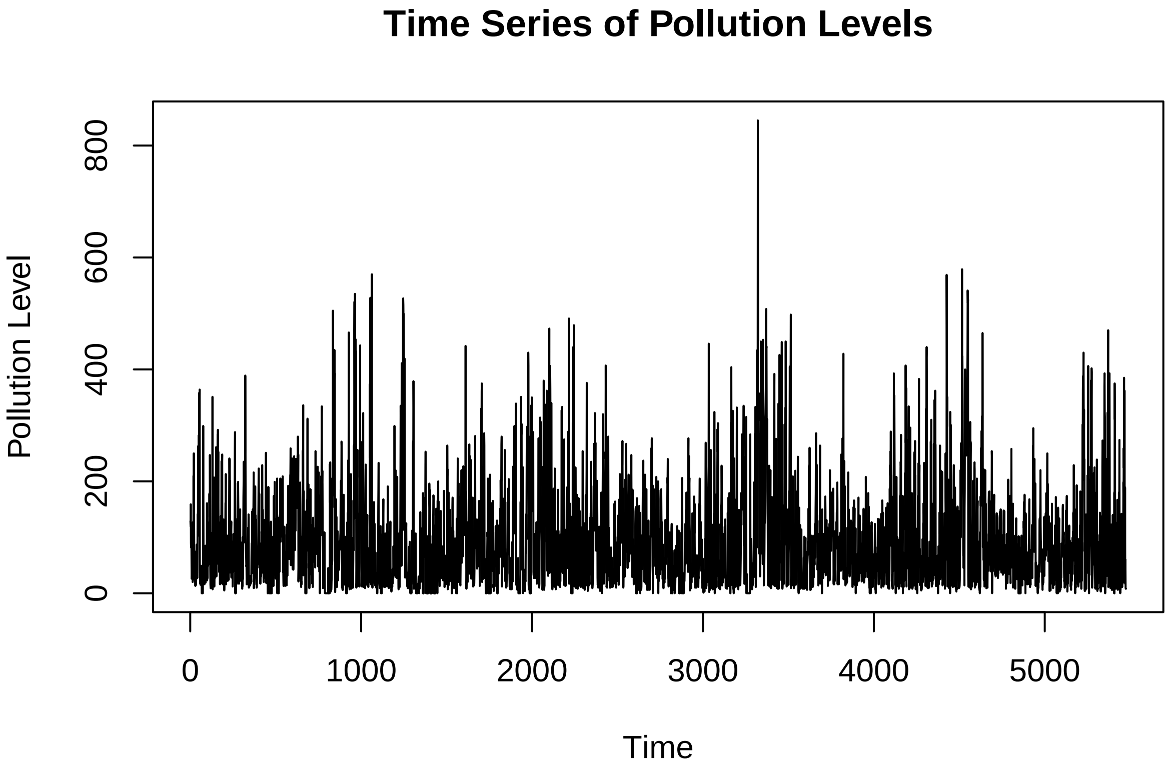
Box-Jenkins approach

```
library(tidyverse)
df <- read.csv("filtered_pollution_data.csv")
```

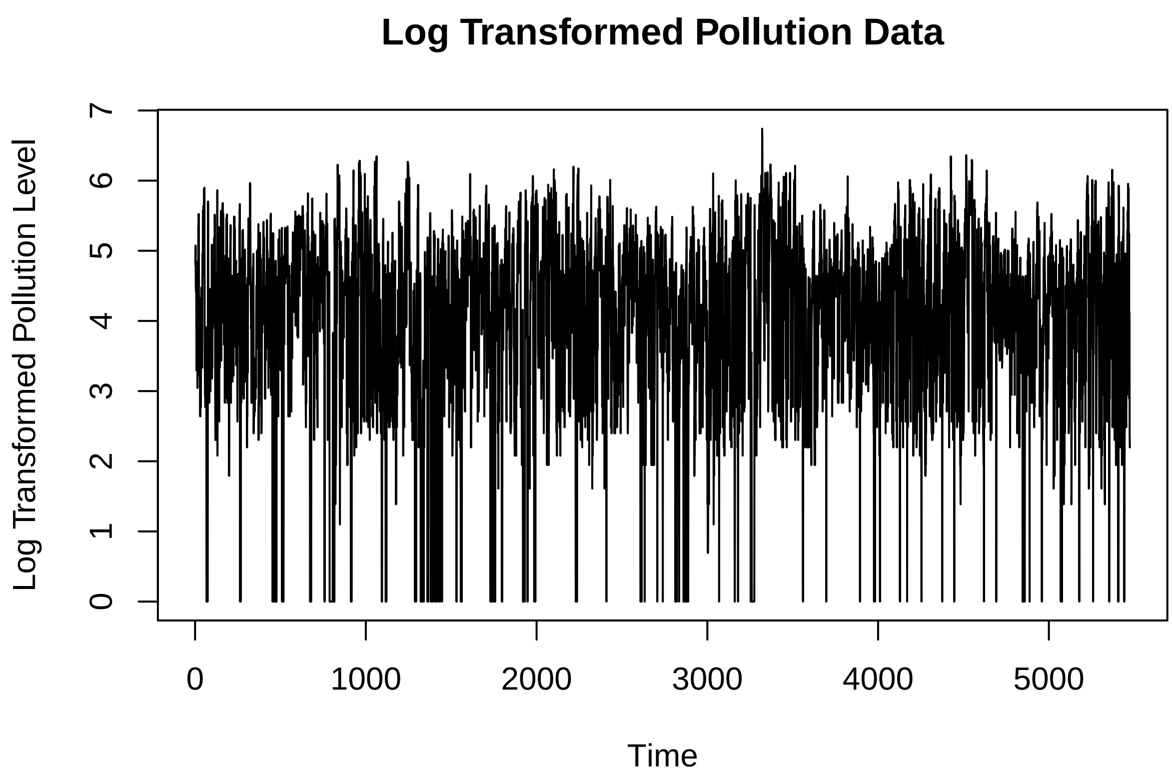
Step 1: Plot the data

```
plot.ts(df$pollution, main="Time Series of Pollution Levels",
        ylab="Pollution Level", xlab="Time")
```



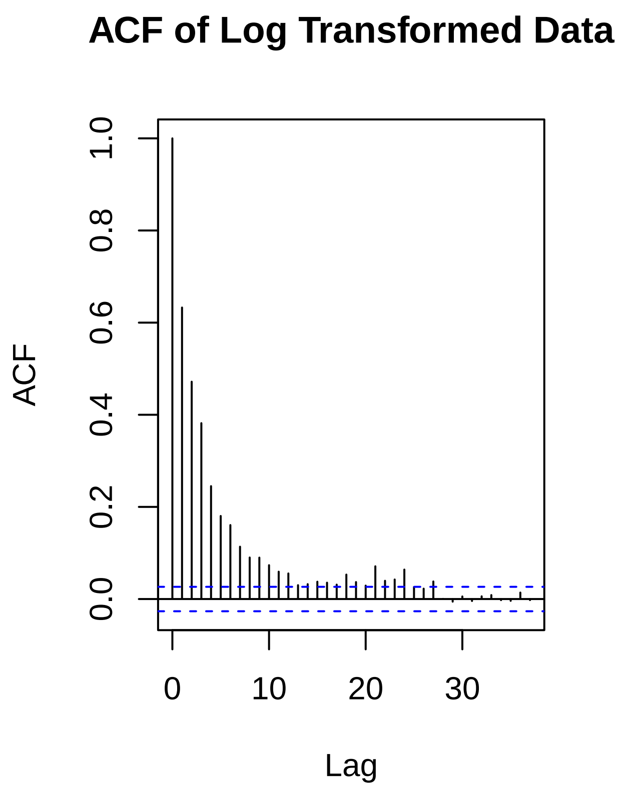
Step 2: Transformation

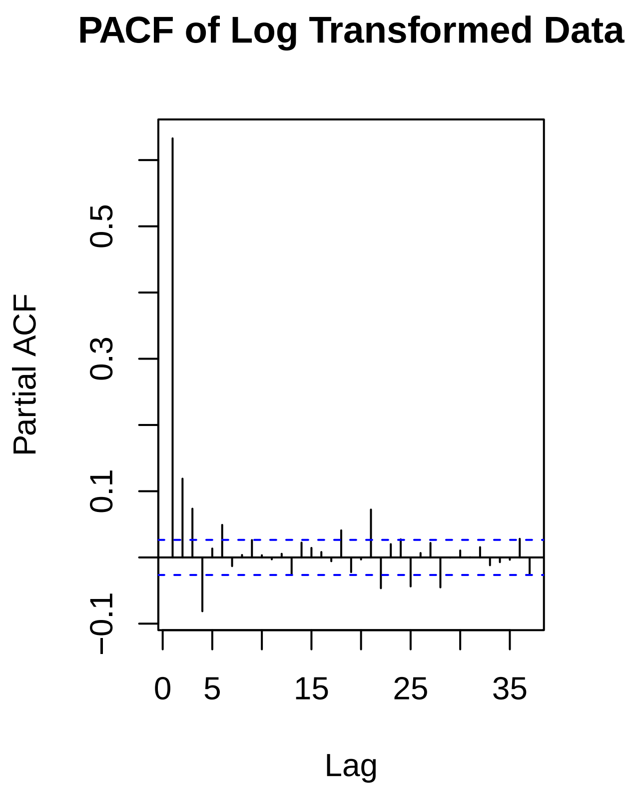
```r
df$pollution_log <- log(df$pollution + 1)

plot.ts(df$pollution_log, main="Log Transformed Pollution Data",
        ylab="Log Transformed Pollution Level", xlab="Time")
```



Step 3: Identification

```r
library(forecast)
library(tseries)
par(mfrow=c(1,2))

acf(df$pollution_log, main="ACF of Log Transformed Data")
pacf(df$pollution_log, main="PACF of Log Transformed Data")
```






```
par(mfrow=c(1,1))

adf.test(df$pollution_log)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  df$pollution_log
## Dickey-Fuller = -14.066, Lag order = 17, p-value = 0.01
## alternative hypothesis: stationary
```

Step 4: Estimation of parameters

```
library(astsa)
```

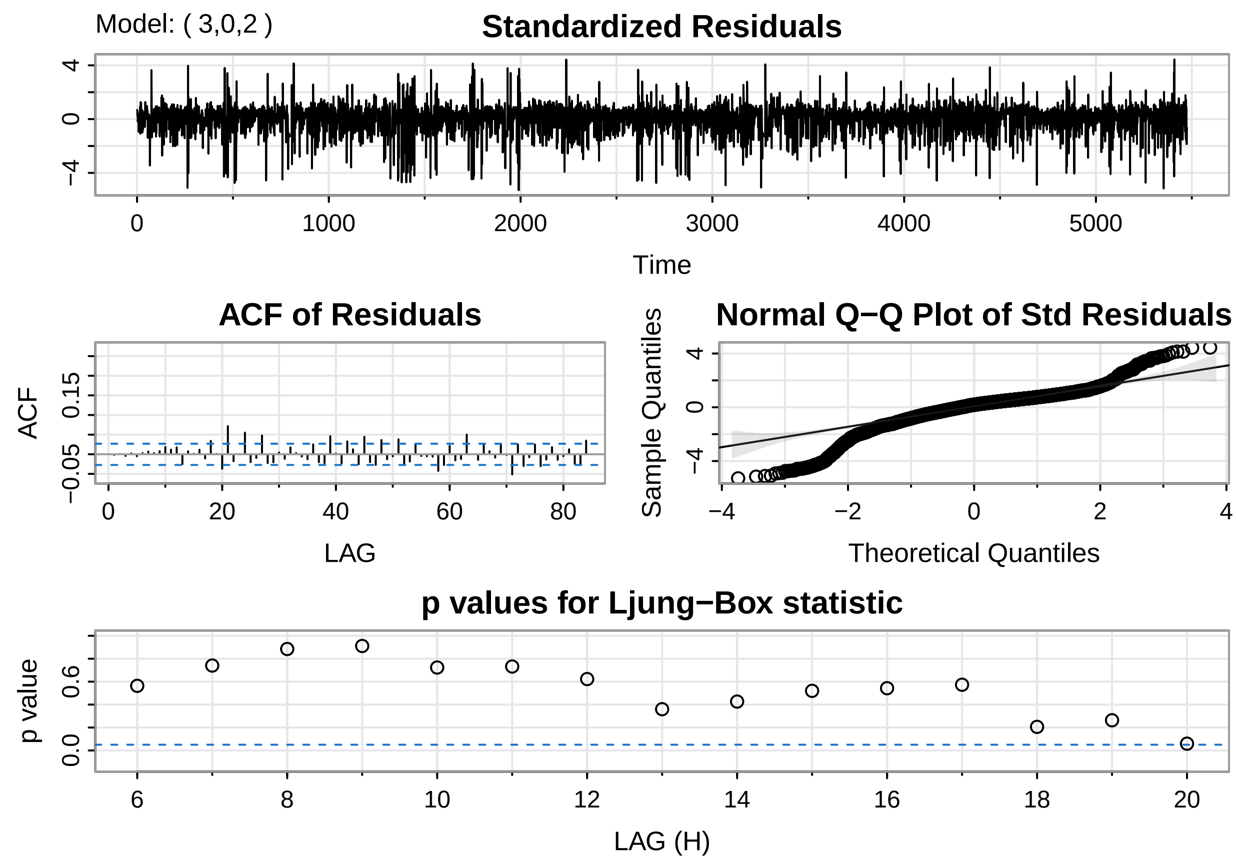
```
##
##    'astsa'
```

```
## The following object is masked from 'package:forecast':
##
##     gas
```

```r
fit <- sarima(df$pollution_log, p=3, d=0, q=2)
```

```
## initial  value 0.283782
## iter   2 value 0.078433
## iter   3 value 0.029221
## iter   4 value 0.017241
## iter   5 value 0.016198
## iter   6 value 0.016088
## iter   7 value 0.016035
## iter   8 value 0.015730
## iter   9 value 0.015095
## iter  10 value 0.013850
## iter  11 value 0.013514
## iter  12 value 0.013398
## iter  13 value 0.013337
## iter  14 value 0.013325
## iter  15 value 0.013324
## iter  16 value 0.013323
## iter  17 value 0.013322
## iter  18 value 0.013319
## iter  19 value 0.013313
## iter  20 value 0.013307
## iter  21 value 0.013306
## iter  22 value 0.013305
## iter  23 value 0.013305
## iter  24 value 0.013305
## iter  25 value 0.013305
## iter  26 value 0.013304
## iter  27 value 0.013304
## iter  28 value 0.013304
## iter  29 value 0.013304
## iter  29 value 0.013304
## iter  29 value 0.013304
## final  value 0.013304
## converged
## initial  value 0.013170
## iter   2 value 0.013170
## iter   3 value 0.013170
## iter   4 value 0.013170
## iter   5 value 0.013170
## iter   6 value 0.013170
## iter   7 value 0.013170
## iter   8 value 0.013170
## iter   8 value 0.013170
## iter   8 value 0.013170
## final  value 0.013170
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##       Estimate     SE t.value p.value
## ar1     0.0623 0.0748  0.8322  0.4053
## ar2    -0.0962 0.0600 -1.6043  0.1087
```

```
## ar3      0.4119 0.0360 11.4350  0.0000
## ma1      0.4959 0.0787  6.2996  0.0000
## ma2      0.4585 0.0602  7.6218  0.0000
## xmean    3.9563 0.0430 91.9885  0.0000
##
## sigma^2 estimated as 1.026576 on 5469 degrees of freedom
##
## AIC = 2.866773  AICc = 2.866776  BIC = 2.875222
##
```



Step 5: Diagnostics

1. There is no obvious patterns and no outliers exceeding 3 standard deviations.
2. The plot of the estimated ACF of the residuals shows nearly no apparent departure from the model assumption, except a few lags.
3. The normal Q-Q plot of the residuals show that the assumption of normality is reasonable, although there are some outliers.
4. The Q-statistic is not statistically significant.

Step 6: Model selection

```
df_best <- auto.arima(df$pollution_log, seasonal = FALSE, ic="bic")
summary(df_best)
```

```
## Series: df$pollution_log
## ARIMA(3,0,2) with non-zero mean
##
## Coefficients:
##          ar1      ar2     ar3     ma1     ma2    mean
##       0.0623  -0.0962  0.4119  0.4959  0.4585  3.9563
## s.e.  0.0748   0.0600  0.0360  0.0787  0.0602  0.0430
##
## sigma^2 = 1.028:  log likelihood = -7840.79
## AIC=15695.58   AICc=15695.6   BIC=15741.84
##
## Training set error measures:
##                          ME     RMSE       MAE  MPE MAPE     MASE        ACF1
## Training set -2.257256e-05 1.013201 0.7233307 -Inf  Inf 1.017692 -0.0018061
```

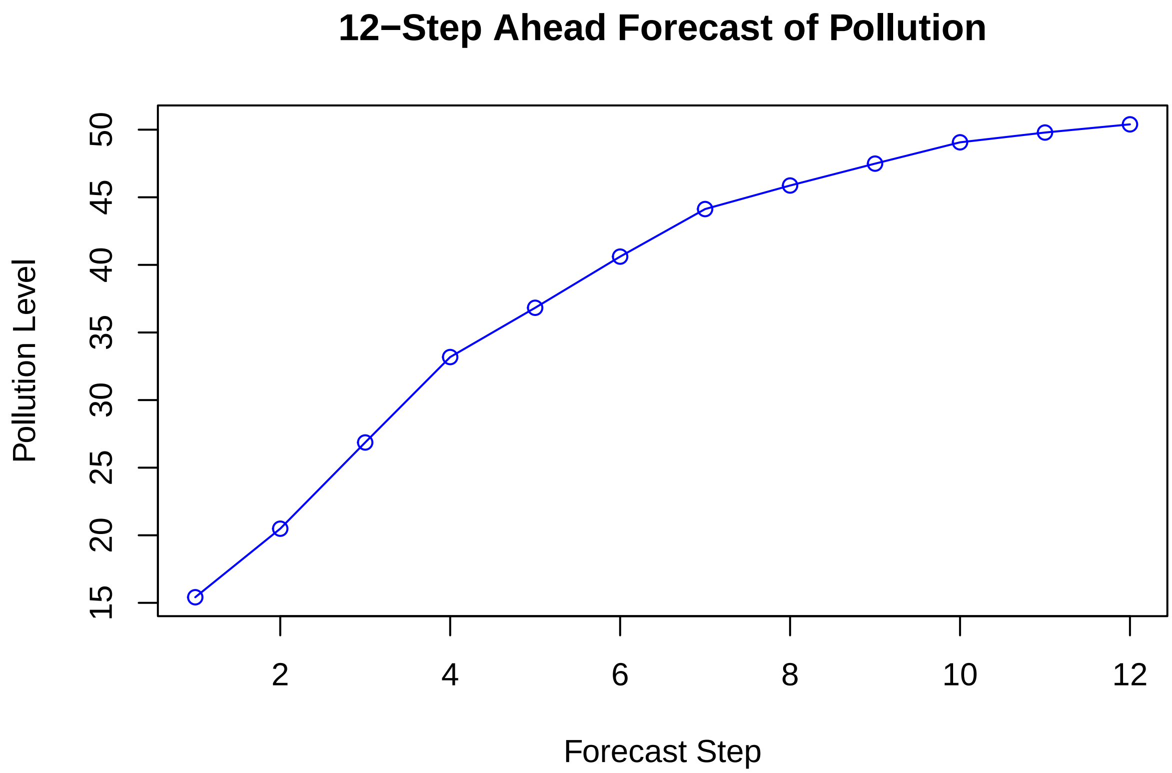
ARIMA(3,0,2) is the best selection for this model.

Additional Step: Forecast

```
df_forecast <- forecast(df_best, h=12)
df_forecast1 <- exp(df_forecast$mean)
df_forecast1
```

```
## Time Series:
## Start = 5476
## End = 5487
## Frequency = 1
##  [1] 15.41574 20.48610 26.86361 33.18135 36.82611 40.61050 44.12829 45.86931
##  [9] 47.48960 49.06628 49.78962 50.39421
```

```
time_index <- 1:12
plot(time_index, df_forecast1,
     type = "o", col = "blue",
     xlab = "Forecast Step",
     ylab = "Pollution Level",
     main = "12-Step Ahead Forecast of Pollution")
```



Unit Root

Step 1: DF Test

```
df_result <- adf.test(df$pollution_log, k = 0)
```

```
## Warning in adf.test(df$pollution_log, k = 0): p-value smaller than printed p-
## value
```

```
print(df_result)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  df$pollution_log
## Dickey-Fuller = -35.097, Lag order = 0, p-value = 0.01
## alternative hypothesis: stationary
```

Step 2: ADF Test

```r
adf_result <- adf.test(df$pollution_log)
```

```
## Warning in adf.test(df$pollution_log): p-value smaller than printed p-value
```

```r
print(adf_result)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  df$pollution_log
## Dickey-Fuller = -14.066, Lag order = 17, p-value = 0.01
## alternative hypothesis: stationary
```

Step 3: PP Test

```r
pp_result <- pp.test(df$pollution_log)
```

```
## Warning in pp.test(df$pollution_log): p-value smaller than printed p-value
```

```r
print(pp_result)
```

```
##
##  Phillips-Perron Unit Root Test
##
## data:  df$pollution_log
## Dickey-Fuller Z(alpha) = -2257.2, Truncation lag parameter = 10,
## p-value = 0.01
## alternative hypothesis: stationary
```

ARMAX Method

Selection criteria

```r
library(vars)
dfvars <- cbind(pollution_log = df$pollution_log,
                   TEMP = df$temp,
                   DEW = df$dew,
                   PRESS = df$press)

dfvars <- na.omit(dfvars)
VARselect(dfvars, lag.max = 10, type = "both")
```

```
## $selection
## AIC(n)  HQ(n)  SC(n) FPE(n)
##     10     10     10     10
```

```
##
## $criteria
##                1           2          3          4          5          6
## AIC(n)    7.445296    7.022673   6.006787   5.848057   5.785693   5.722004
## HQ(n)     7.455418    7.039543   6.030406   5.878424   5.822809   5.765868
## SC(n)     7.474307    7.071025   6.074480   5.935091   5.892068   5.847720
## FPE(n) 1711.791789 1121.780740 406.176336 346.560403 325.607721 305.516801
##                7          8          9         10
## AIC(n)   5.681481   5.653553   5.626481   5.606281
## HQ(n)    5.732092   5.710913   5.690589   5.677138
## SC(n)    5.826537   5.817950   5.810219   5.809360
## FPE(n) 293.383641 285.303468 277.683444 272.130677
```

```r
dfarmax <- VAR(dfvars, p = 2, type = "both")
summary(dfarmax)
```

```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: pollution_log, TEMP, DEW, PRESS
## Deterministic variables: both
## Sample size: 5473
## Log Likelihood: -50244.35
## Roots of the characteristic polynomial:
## 0.9859 0.8228 0.6635 0.6635 0.2761 0.2761 0.2634 0.2098
## Call:
## VAR(y = dfvars, p = 2, type = "both")
##
##
## Estimation results for equation pollution_log:
## ==============================================
## pollution_log = pollution_log.l1 + TEMP.l1 + DEW.l1 + PRESS.l1 + pollution_log.l2 + TEMP.l2 + DEW.l2
##
##                   Estimate Std. Error t value Pr(>|t|)
## pollution_log.l1  5.166e-01  1.466e-02  35.246  < 2e-16 ***
## TEMP.l1           7.685e-03  3.589e-03   2.141  0.03228 *
## DEW.l1            1.640e-02  3.930e-03   4.173 3.05e-05 ***
## PRESS.l1         -6.353e-02  5.968e-03 -10.645  < 2e-16 ***
## pollution_log.l2  1.693e-01  1.448e-02  11.693  < 2e-16 ***
## TEMP.l2           1.045e-02  3.503e-03   2.983  0.00286 **
## DEW.l2           -2.416e-02  4.000e-03  -6.041 1.64e-09 ***
## PRESS.l2          7.530e-02  5.927e-03  12.704  < 2e-16 ***
## const            -1.095e+01  2.697e+00  -4.060 4.97e-05 ***
## trend             6.133e-06  8.597e-06   0.713  0.47560
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.9934 on 5463 degrees of freedom
## Multiple R-Squared: 0.4416,  Adjusted R-squared: 0.4407
## F-statistic:   480 on 9 and 5463 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation TEMP:
```

```
## ======================================
## TEMP = pollution_log.l1 + TEMP.l1 + DEW.l1 + PRESS.l1 + pollution_log.l2 + TEMP.l2 + DEW.l2 + PRESS.l
##
##                    Estimate Std. Error t value Pr(>|t|)
## pollution_log.l1 -3.158e-01  7.269e-02  -4.345 1.42e-05 ***
## TEMP.l1           4.818e-01  1.780e-02  27.073  < 2e-16 ***
## DEW.l1            3.767e-01  1.949e-02  19.331  < 2e-16 ***
## PRESS.l1          4.536e-01  2.960e-02  15.326  < 2e-16 ***
## pollution_log.l2 -1.460e-01  7.179e-02  -2.033   0.0421 *
## TEMP.l2          -3.423e-02  1.737e-02  -1.971   0.0488 *
## DEW.l2           -3.135e-02  1.984e-02  -1.580   0.1141
## PRESS.l2         -6.038e-01  2.939e-02 -20.541  < 2e-16 ***
## const             1.603e+02  1.337e+01  11.985  < 2e-16 ***
## trend             1.726e-04  4.263e-05   4.050 5.19e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 4.926 on 5463 degrees of freedom
## Multiple R-Squared: 0.8403,  Adjusted R-squared:  0.84
## F-statistic:  3194 on 9 and 5463 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation DEW:
## =====================================
## DEW = pollution_log.l1 + TEMP.l1 + DEW.l1 + PRESS.l1 + pollution_log.l2 + TEMP.l2 + DEW.l2 + PRESS.l2
##
##                    Estimate Std. Error t value Pr(>|t|)
## pollution_log.l1  2.813e-01  5.534e-02   5.082 3.85e-07 ***
## TEMP.l1           8.084e-02  1.355e-02   5.966 2.58e-09 ***
## DEW.l1            6.547e-01  1.484e-02  44.124  < 2e-16 ***
## PRESS.l1         -4.136e-01  2.253e-02 -18.353  < 2e-16 ***
## pollution_log.l2 -1.725e-01  5.466e-02  -3.156  0.00161 **
## TEMP.l2           2.451e-01  1.323e-02  18.530  < 2e-16 ***
## DEW.l2            1.244e-01  1.510e-02   8.235 2.23e-16 ***
## PRESS.l2          5.142e-01  2.238e-02  22.975  < 2e-16 ***
## const            -1.061e+02  1.018e+01 -10.420  < 2e-16 ***
## trend            -9.896e-05  3.246e-05  -3.049  0.00231 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 3.75 on 5463 degrees of freedom
## Multiple R-Squared: 0.9327,  Adjusted R-squared: 0.9326
## F-statistic:  8418 on 9 and 5463 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation PRESS:
## =======================================
## PRESS = pollution_log.l1 + TEMP.l1 + DEW.l1 + PRESS.l1 + pollution_log.l2 + TEMP.l2 + DEW.l2 + PRESS
##
##                    Estimate Std. Error t value Pr(>|t|)
## pollution_log.l1  1.133e-01  4.409e-02   2.569  0.01021 *
## TEMP.l1           1.300e-01  1.080e-02  12.037  < 2e-16 ***
```

```
## DEW.l1           -1.725e-01  1.182e-02 -14.590  < 2e-16 ***
## PRESS.l1          1.044e+00  1.795e-02  58.168  < 2e-16 ***
## pollution_log.l2  1.320e-01  4.355e-02   3.032  0.00244 **
## TEMP.l2          -5.147e-04  1.054e-02  -0.049  0.96104
## DEW.l2            3.692e-02  1.203e-02   3.068  0.00217 **
## PRESS.l2         -1.108e-01  1.783e-02  -6.214 5.54e-10 ***
## const             6.532e+01  8.113e+00   8.052 9.92e-16 ***
## trend            -3.753e-05  2.586e-05  -1.451  0.14672
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 2.988 on 5463 degrees of freedom
## Multiple R-Squared: 0.9157,  Adjusted R-squared: 0.9156
## F-statistic:  6595 on 9 and 5463 DF,  p-value: < 2.2e-16
##
##
##
## Covariance matrix of residuals:
##               pollution_log     TEMP     DEW    PRESS
## pollution_log        0.9868  -0.5648  1.5559  -0.2490
## TEMP                -0.5648  24.2654 -4.4531 -10.1700
## DEW                  1.5559  -4.4531 14.0662  -0.2167
## PRESS               -0.2490 -10.1700 -0.2167   8.9297
##
## Correlation matrix of residuals:
##               pollution_log    TEMP      DEW    PRESS
## pollution_log       1.00000 -0.1154  0.41763 -0.08389
## TEMP               -0.11542  1.0000 -0.24104 -0.69089
## DEW                 0.41763 -0.2410  1.00000 -0.01934
## PRESS              -0.08389 -0.6909 -0.01934  1.00000
```

```r
dfarmax <- VAR(dfvars, p = 2, type = "both")
summary(dfarmax)
```

```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: pollution_log, TEMP, DEW, PRESS
## Deterministic variables: both
## Sample size: 5473
## Log Likelihood: -50244.35
## Roots of the characteristic polynomial:
## 0.9859 0.8228 0.6635 0.6635 0.2761 0.2761 0.2634 0.2098
## Call:
## VAR(y = dfvars, p = 2, type = "both")
##
##
## Estimation results for equation pollution_log:
## ==============================================
## pollution_log = pollution_log.l1 + TEMP.l1 + DEW.l1 + PRESS.l1 + pollution_log.l2 + TEMP.l2 + DEW.l2
##
##                     Estimate Std. Error t value Pr(>|t|)
## pollution_log.l1  5.166e-01  1.466e-02  35.246  < 2e-16 ***
```

```
## TEMP.l1           7.685e-03  3.589e-03   2.141  0.03228 *
## DEW.l1            1.640e-02  3.930e-03   4.173 3.05e-05 ***
## PRESS.l1         -6.353e-02  5.968e-03 -10.645  < 2e-16 ***
## pollution_log.l2  1.693e-01  1.448e-02  11.693  < 2e-16 ***
## TEMP.l2           1.045e-02  3.503e-03   2.983  0.00286 **
## DEW.l2           -2.416e-02  4.000e-03  -6.041 1.64e-09 ***
## PRESS.l2          7.530e-02  5.927e-03  12.704  < 2e-16 ***
## const            -1.095e+01  2.697e+00  -4.060 4.97e-05 ***
## trend             6.133e-06  8.597e-06   0.713  0.47560
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.9934 on 5463 degrees of freedom
## Multiple R-Squared: 0.4416,  Adjusted R-squared: 0.4407
## F-statistic:   480 on 9 and 5463 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation TEMP:
## =====================================
## TEMP = pollution_log.l1 + TEMP.l1 + DEW.l1 + PRESS.l1 + pollution_log.l2 + TEMP.l2 + DEW.l2 + PRESS.l
##
##                    Estimate Std. Error t value Pr(>|t|)
## pollution_log.l1 -3.158e-01  7.269e-02  -4.345 1.42e-05 ***
## TEMP.l1           4.818e-01  1.780e-02  27.073  < 2e-16 ***
## DEW.l1            3.767e-01  1.949e-02  19.331  < 2e-16 ***
## PRESS.l1          4.536e-01  2.960e-02  15.326  < 2e-16 ***
## pollution_log.l2 -1.460e-01  7.179e-02  -2.033   0.0421 *
## TEMP.l2          -3.423e-02  1.737e-02  -1.971   0.0488 *
## DEW.l2           -3.135e-02  1.984e-02  -1.580   0.1141
## PRESS.l2         -6.038e-01  2.939e-02 -20.541  < 2e-16 ***
## const             1.603e+02  1.337e+01  11.985  < 2e-16 ***
## trend             1.726e-04  4.263e-05   4.050 5.19e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 4.926 on 5463 degrees of freedom
## Multiple R-Squared: 0.8403,  Adjusted R-squared:  0.84
## F-statistic:  3194 on 9 and 5463 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation DEW:
## ====================================
## DEW = pollution_log.l1 + TEMP.l1 + DEW.l1 + PRESS.l1 + pollution_log.l2 + TEMP.l2 + DEW.l2 + PRESS.l2
##
##                    Estimate Std. Error t value Pr(>|t|)
## pollution_log.l1  2.813e-01  5.534e-02   5.082 3.85e-07 ***
## TEMP.l1           8.084e-02  1.355e-02   5.966 2.58e-09 ***
## DEW.l1            6.547e-01  1.484e-02  44.124  < 2e-16 ***
## PRESS.l1         -4.136e-01  2.253e-02 -18.353  < 2e-16 ***
## pollution_log.l2 -1.725e-01  5.466e-02  -3.156  0.00161 **
## TEMP.l2           2.451e-01  1.323e-02  18.530  < 2e-16 ***
## DEW.l2            1.244e-01  1.510e-02   8.235 2.23e-16 ***
```

```
## PRESS.l2          5.142e-01  2.238e-02  22.975  < 2e-16 ***
## const            -1.061e+02  1.018e+01 -10.420  < 2e-16 ***
## trend            -9.896e-05  3.246e-05  -3.049  0.00231 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 3.75 on 5463 degrees of freedom
## Multiple R-Squared: 0.9327,  Adjusted R-squared: 0.9326
## F-statistic:  8418 on 9 and 5463 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation PRESS:
## ======================================
## PRESS = pollution_log.l1 + TEMP.l1 + DEW.l1 + PRESS.l1 + pollution_log.l2 + TEMP.l2 + DEW.l2 + PRESS
##
##                    Estimate Std. Error t value Pr(>|t|)
## pollution_log.l1  1.133e-01  4.409e-02   2.569  0.01021 *
## TEMP.l1           1.300e-01  1.080e-02  12.037  < 2e-16 ***
## DEW.l1           -1.725e-01  1.182e-02 -14.590  < 2e-16 ***
## PRESS.l1          1.044e+00  1.795e-02  58.168  < 2e-16 ***
## pollution_log.l2  1.320e-01  4.355e-02   3.032  0.00244 **
## TEMP.l2          -5.147e-04  1.054e-02  -0.049  0.96104
## DEW.l2            3.692e-02  1.203e-02   3.068  0.00217 **
## PRESS.l2         -1.108e-01  1.783e-02  -6.214 5.54e-10 ***
## const             6.532e+01  8.113e+00   8.052 9.92e-16 ***
## trend            -3.753e-05  2.586e-05  -1.451  0.14672
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 2.988 on 5463 degrees of freedom
## Multiple R-Squared: 0.9157,  Adjusted R-squared: 0.9156
## F-statistic:  6595 on 9 and 5463 DF,  p-value: < 2.2e-16
##
##
##
## Covariance matrix of residuals:
##              pollution_log     TEMP     DEW    PRESS
## pollution_log       0.9868  -0.5648  1.5559  -0.2490
## TEMP               -0.5648  24.2654 -4.4531 -10.1700
## DEW                 1.5559  -4.4531 14.0662  -0.2167
## PRESS              -0.2490 -10.1700 -0.2167   8.9297
##
## Correlation matrix of residuals:
##              pollution_log    TEMP      DEW    PRESS
## pollution_log      1.00000 -0.1154  0.41763 -0.08389
## TEMP              -0.11542  1.0000 -0.24104 -0.69089
## DEW                0.41763 -0.2410  1.00000 -0.01934
## PRESS             -0.08389 -0.6909 -0.01934  1.00000
```

Residual ACF/CCF and Prediction

```r
serial.test(dfarmax, lags.pt = 12, type = "PT.adjusted")
```

```
##
##  Portmanteau Test (adjusted)
##
## data:  Residuals of VAR object dfarmax
## Chi-squared = 12950, df = 160, p-value < 2.2e-16
```

```r
dfpredict <- predict(dfarmax, n.ahead = 24, ci = 0.95)
fanchart(dfpredict)
```